CSE 429

Technical Writing and Presentation

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Introduction to LaTeX

What is LaTeX?

- A typesetting language, think of a programming language to make documents beautiful. (mathematically related research papers, especially).
- Microsoft Word is built using **"what you see is what you get"**
- LaTeX is written in plain text with **markup tagging.**
- LaTeX is pronounced "lay-tech" or "lah-tech," not "la-teks."

Example

```latex
\documentclass{article} % Starts an article
\usepackage{amsmath} % Imports amsmath
\title{\LaTeX} % Title

\begin{document} % Begins a document
  \maketitle
  \LaTeX{} is a document preparation system for
  the \TeX{} typesetting program. It offers
  programmable desktop publishing features and
  extensive facilities for automating most
  aspects of typesetting and desktop publishing,
  including numbering and  cross-referencing,
  tables and figures, page layout,
  bibliographies, and much more. \LaTeX{} was
  originally written in 1984 by Leslie Lamport
  and has become the  dominant method for using
  \TeX; few people write in plain \TeX{} anymore.
  The current version is \LaTeXe.

  % This is a comment, not shown in final output.
  % The following shows typesetting  power of LaTeX:
  \begin{align}
    E_0 &= mc^2 \\
    E &= \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}
  \end{align}
\end{document}
```

LaTeX

LaTeX is a document preparation system for the TeX typesetting program. It offers programmable desktop publishing features and extensive facilities for automating most aspects of typesetting and desktop publishing, including numbering and cross-referencing, tables and figures, page layout, bibliographies, and much more. LaTeX was originally written in 1984 by Leslie Lamport and has become the dominant method for using TeX; few people write in plain TeX anymore. The current version is LaTeX 2_ε.

$$E_0 = mc^2 \tag{1}$$

$$E = \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}} \tag{2}$$

LaTeX Editors

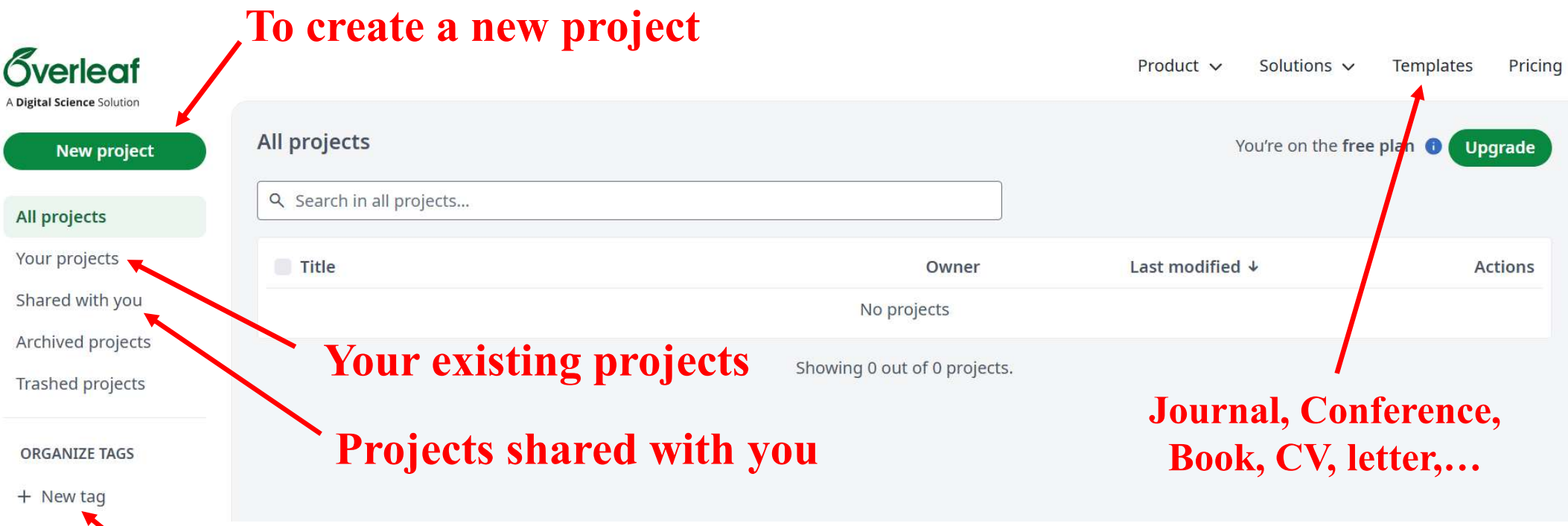
- **Offline:**

  MiKTeX, TeXstudio, TeX Live, TeXworks, TeXMaker, TeXnicCenter….. Etc.

- **Online:**

  Overleaf, ShareLaTeX, Papeeria, LaTeXBase, LaTeXLab,……. etc.

Overleaf interface



To create a new project

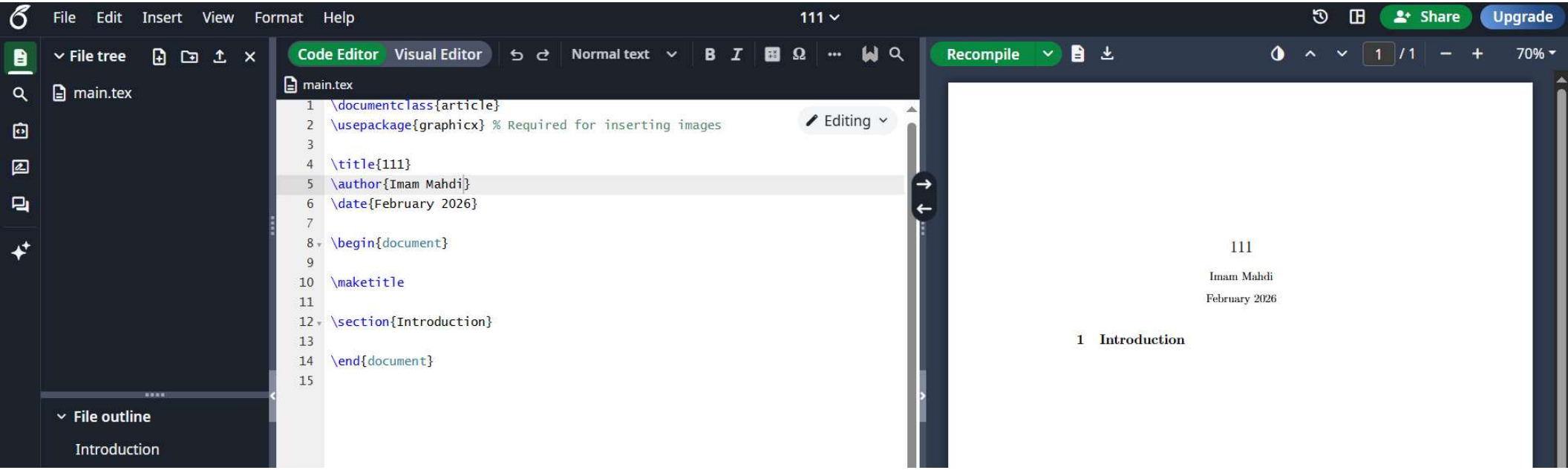
Your existing projects

Projects shared with you

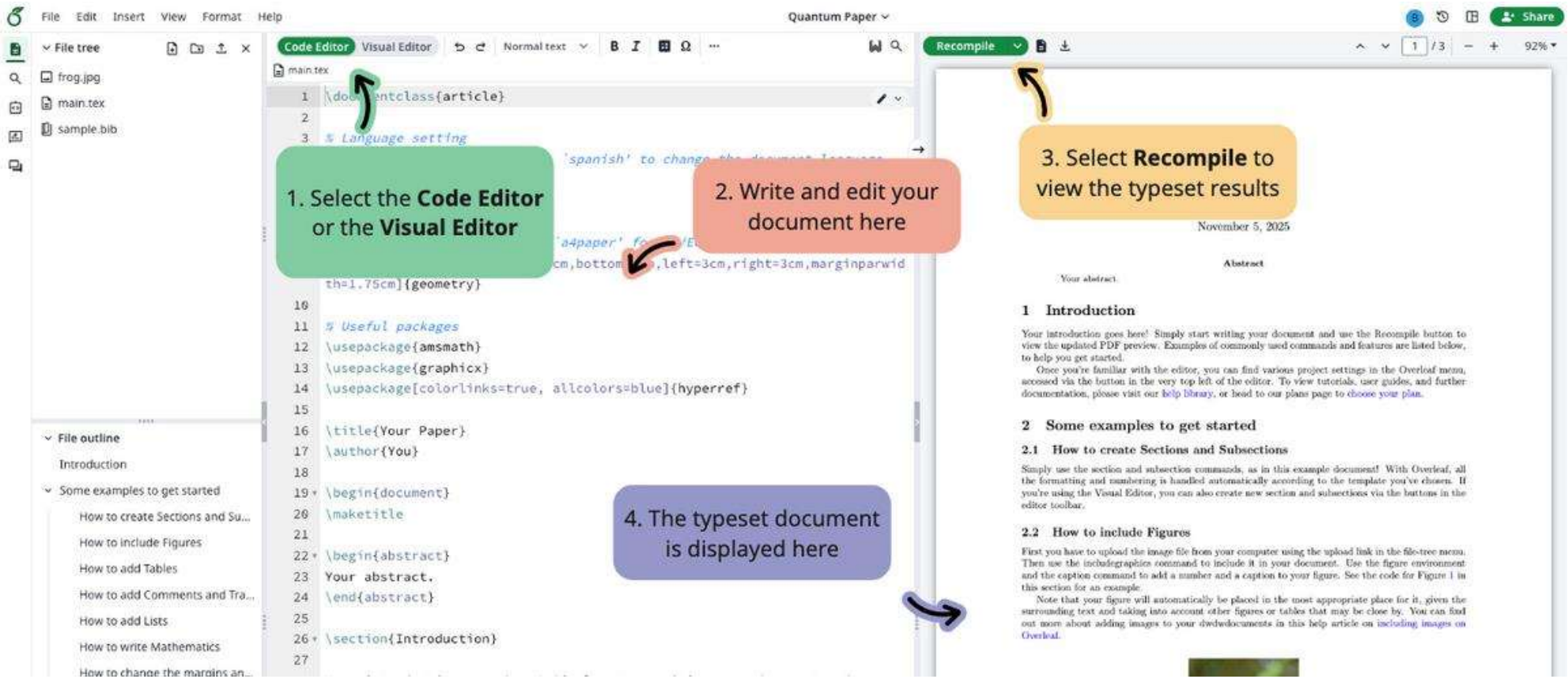
Journal, Conference, Book, CV, letter,…

To classify projects (Folder)

LaTeX interface Overview



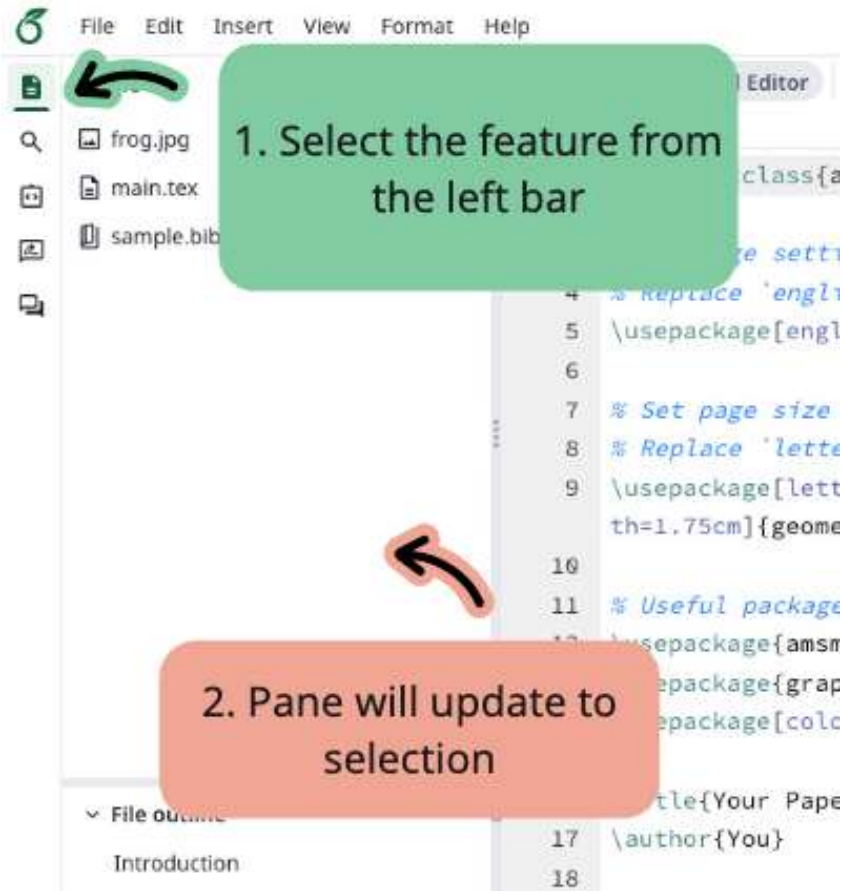
LaTeX interface Overview



LaTeX interface Overview

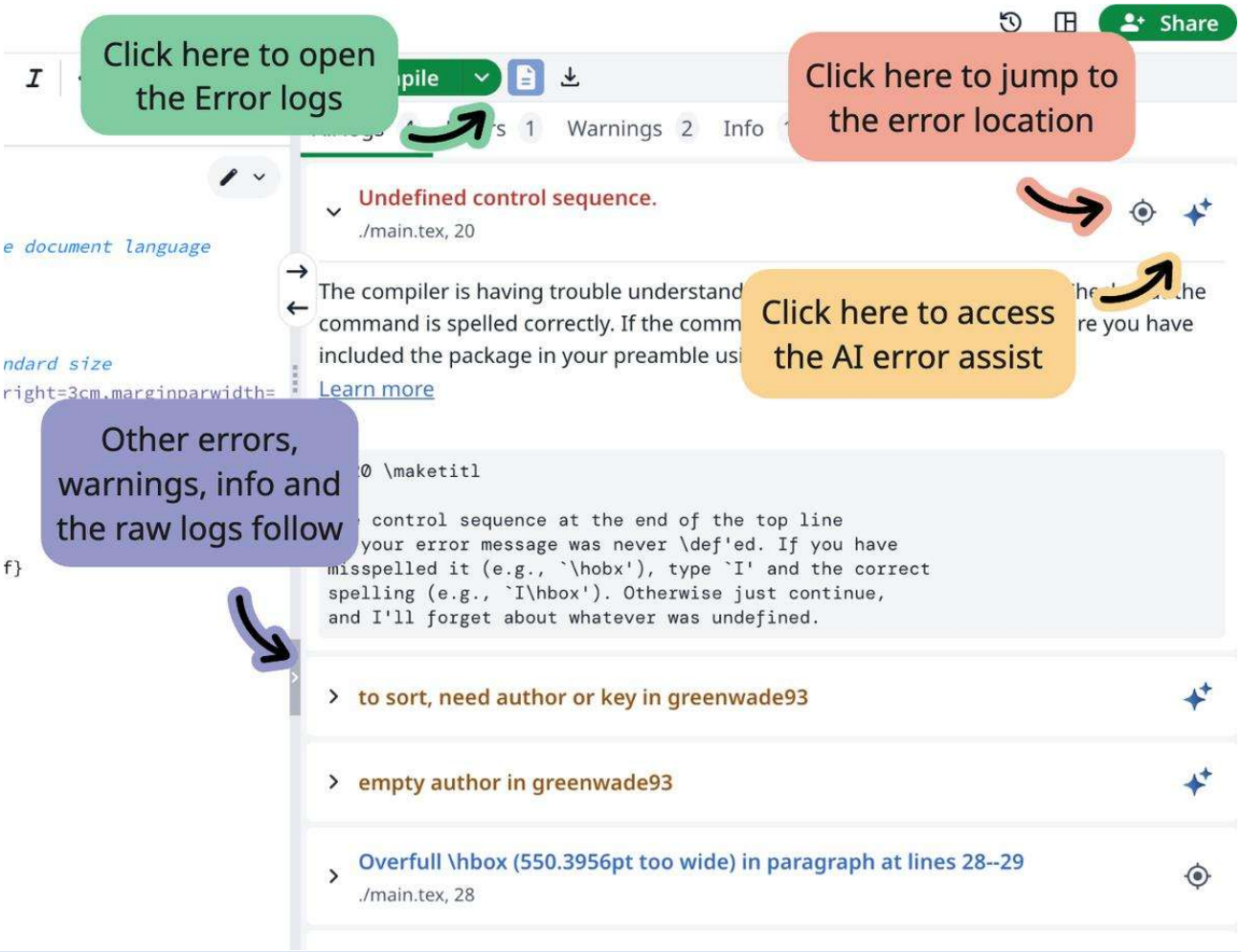
- **Overleaf Icon**: Clicking the Overleaf icon in the top-left corner will now take you back to your project dashboard.
- **Menus**: We've introduced familiar menu options like File, Edit, View, and Help to make common actions easier to find.
- **Project menu**: Clicking the drop-down menu next to the project name gives you access to project-level actions, such as downloading the source, renaming the project, making a copy, or submitting to a journal.
- **Top-Right Controls**: On the top-right, you'll find buttons for History, Share, and options to change the editor layout.

LaTeX interface Overview



- **File tree:** Organize and manage your project's files and directories.
- **Search:** Quickly locate information within your project files.
- **Integrations:** Connect with external platforms, such as Papers and GitHub, to enhance workflow.
- **Review:** Utilize features like Track Changes and Comments for collaborative editing.
- **Chat:** Communicate in real-time with team members or collaborators.

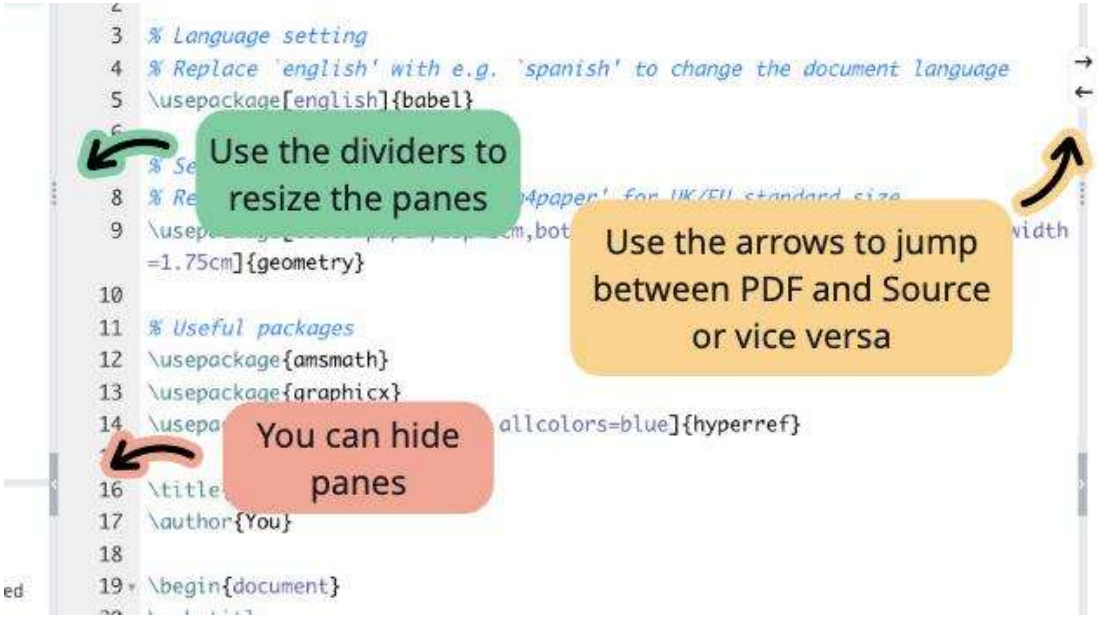
LaTeX interface Overview



- The error logs remain next to the Recompile button with a new design. Clicking the icon will open a pane displaying any LaTeX errors, warnings, and other compiler output to help you debug your project. The AI Error Assistant and other log-related options are still available in this view.

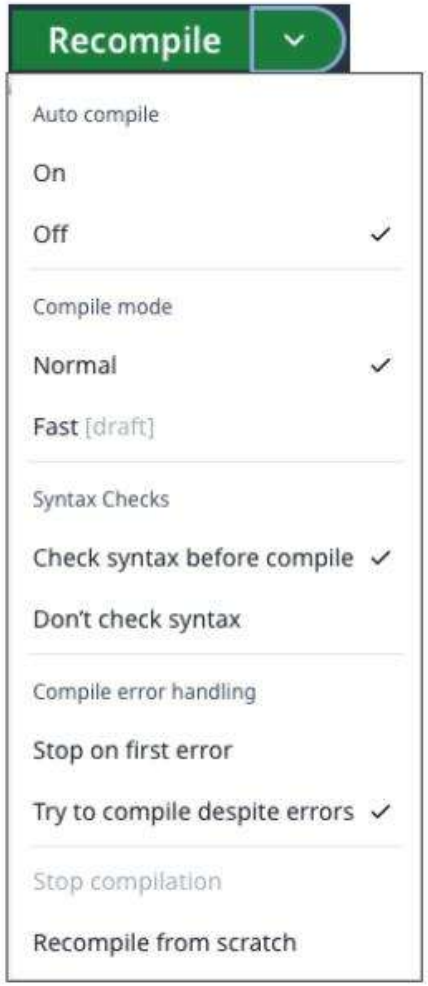
- As before, whenever there are errors, solve errors from the top down, resolve them as soon as possible, and check the logs regularly.

LaTeX interface Overview



- You can still use the dividers to resize or hide panes. The arrows for SyncTeX (to jump between the source code and the PDF) are still located on the divider between the editor and PDF viewer.

LaTeX interface Overview



- **Auto compile**
  When this option is turned on, Overleaf automatically compiles your document for you every few seconds.
- **Compile Mode:**
  The LaTeX compiler can be given an instruction to produce a fast draft of the PDF, rather than a complete document. The main effect of choosing the Fast [draft] option is to skip over image file processing.
- **Syntax Checks:**
  Overleaf automatically checks your LaTeX code for common syntax issues as you type. These checks help catch typos and errors that could prevent your project from compiling properly.
- **Compile error handling:**
  The default mode for LaTeX compilation on Overleaf is for the compiler to attempt to produce a PDF even if there are errors. In this mode, the compiler will ignore or compensate for errors it encounters when compiling.

LaTeX interface Overview

- **For detailed information, go through this link:**

  https://docs.overleaf.com/

Basic Document Structure

- The format of a document is pretty simple:

➢ **In the preamble:**
Documentclass
Packages

➢ **In the front matter:**
Title / Author

➢ **In the body:**
Contents

➢ **In the back matter:**
bibliography

In the Preamble

- You specify your document class
  Document classes:

  \documentclass[12pt]{article}
  \documentclass{article}

Class	Best used for
Article	Short reports, journal articles
Report	Thesis, long reports
Book	Books, long documents
Beamer	Presentation
Letter	Letters
memoir	Complex books, theses

- **Packages:**
  Give the programming language the ability to do certain things

  \usepackage[margin=1in]{geometry}
- **\usepackage{setspace}**
- **\usepackage{harvard}**

Latex Command

- LaTeX commands begin with a **backslash**
- Calling a command looks like this:

\command[optional argument]{argument}

- Arguments in curly braces are mandatory

```
\documentclass[12pt]{article}
\usepackage[margin=1in]{geometry}
% setting margins
\usepackage{setspace}
% nice line spacing package
\usepackage{harvard}
% family of seven bibliography styles including APSR
\bibliographystyle{apsr}
```

In the front matter

- \title{}
- \author{}
- \begin{document}
- \maketitle
- \begin{abstract}

  \end{abstract}
- \pagebreak

```
\begin{document}
\title{Writing My First \LaTeX Article}
\author{Byungwon Woo}

\maketitle
\singlespacing

\begin{abstract}
\noindent My first article written in LaTeX(or \LaTeX) using WinEdt.
\end{abstract}
```

In the body

- To begin a new section

  \section{}

  Similarly, **\subsection{}, \subsubsection{}, \subsubsubsection{}**
- LaTeX does automatic numbering. If you don't like it, use **section*{}**
- \emph{}, \textbf{}
- \singlespacing, \doublespacing, \onehalfspacing
- \centering or \begin{centering} & \end{centering}
- Tables, equations, figures…… etc.

In the back matter

- Don't forget **bibliography{filename}**

  Make sure that the **BibTeX** file is saved in the same location where the main TeX file is saved.

- Don't forget **end{document}**

Spaces, Line breaks, Paragraph & Comments

- LaTeX treats **multiple spaces (in a line) just like a single space**.
- A single line break has the same effect as a single space.
- Multiple empty lines are treated like one.
- Use **\\ (double backslash)** for a newline or **\hfill \break**.

  Hit **return twice** for a new paragraph.

- Everything following a percent sign (**%** symbol used as comment) until the end of the line.

Horizontal blank spaces

- **\hspace{1cm}**

  Inserts a horizontal space whose length is 1cm.

- **\hfill**

  Inserts a blank space that will stretch accordingly to fill the space available.

Horizontal spaces can be inserted manually. Useful to control the fine-tuning in the layout of pictures.

Left Side Right Side

Vertical blank spaces

- **\vspace{5mm}**

  Inserts a vertical spaces whose length is 5mm. Other LaTeX units can be used with this command.

- **\vfill**

  Inserts a blank space that will stretch accordingly to fill the vertical space available.

Lab Task 2

Paragraph Title

Your Name, ID: , Section:

Date:

Write a paragraph on "Aim in Life" (Fictional ideas are allowed). Use the line break, new line commands, and add comments.

Write at least 1page.

In the lab report, add: Cover page, Software, Describe the commands used in the whole code.

Result: Give a screenshot of the code and merge the PDF result part.

Challenges (you've faced regarding the task).

Thank You